

AI-Assisted Mixed-Signal Physical Design Flow

Neha Pal

Week 1 Task Report – Mixed-Signal Physical Design

Abstract—This report summarizes Week 1 progress on the VSD mixed-signal physical design flow. It includes installation and verification of OpenROAD, OpenLane, and Magic VLSI, along with study of repository structure and analog macro integration. The AMUX2_3V layout was analyzed in Magic VLSI, and a DRC experiment was performed by introducing and correcting a layout violation. The work provided hands-on exposure to physical design flow, layout verification, and mixed-signal integration.

I. INTRODUCTION

The objective of this task was to understand the complete mixed-signal physical design flow using the vsdmixedsignalflow repository. The study focused on tool installation, repository structure analysis, analog macro understanding, OpenLane execution, Magic layout visualization, DRC analysis, and debugging methodology.

II. REPOSITORY UNDERSTANDING

The repo demonstrates open-source mixed-signal physical design flow using OpenLane, OpenROAD, Magic VLSI, and SKY130 PDK. It is built around a digital top-level design (design_mux) that integrates the AMUX2_3V analog macro into a standard RTL-to-GDSII flow. The repository provides different abstraction views of the macro, including LEF for physical representation, LIB for timing and power characterization, and Verilog as a blackbox for synthesis. These abstractions allow seamless integration of the analog block without exposing its internal transistor-level design. The OpenLane flow performs synthesis, floorplanning, placement, routing, and final GDS generation. Magic VLSI is used for layout inspection and design rule checking of the analog macro. Overall this repository served as a practical reference for understanding mixed-signal design integration and automated physical design flow execution.

III. AI PROMPTS/WORKFLOW USED

AI tools were used to assist in understanding the mixed-signal physical design flow and resolving setup and debugging issues in OpenLane, OpenROAD, and Magic VLSI. They helped interpret flow stages, DRC concepts, and repository structure, and supported technical documentation writing. Overall, AI acted as a supportive tool for learning and organizing the physical design workflow.

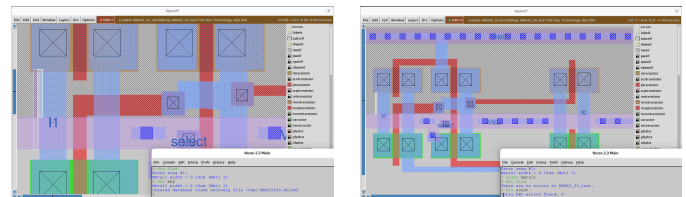
IV. EXPERIMENTS ATTEMPTED

The experiments involved exploring the repository structure and studying key design files such as RTL (design_mux.v) and the AMUX2_3V analog wrapper. LEF, LIB, and OpenLane configuration files were analyzed to understand physical abstraction and flow execution. The layout was visualized in

Magic VLSI, followed by Design Rule Check (DRC) verification. Finally, synthesis, floorplan, placement, and routing outputs were reviewed to understand the complete RTL-to-GDSII flow.

V. RESULTS AND OBSERVATIONS

The experimental flow was successfully executed and was largely consistent with the reference mixed-signal design methodology. The AMUX2_3V analog macro was properly visualized in Magic VLSI, and the top-level design_mux layout was generated with a clean DRC (DRC = 0). To better understand physical verification, a DRC error was intentionally introduced in the layout. This was then corrected to validate the DRC process in a controlled manner. The results confirmed correct integration of analog and digital components within the OpenLane flow. During the process, it was observed that mixed-signal physical design is highly sensitive to correct macro abstraction, including LEF completeness, pin mapping consistency, and tool version compatibility. It was also noted that proper verification requires more than visual inspection, and must include DRC reports and generated flow logs to confirm correctness of the final layout.



(a) Intentional DRC Violation (b) Corrected Layout (DRC = 0)

Fig. 1: AMUX2_3V analog macro layout viewed.

VI. CHALLENGES AND FUTURE WORK

faced challenges in understanding the repository structure and how the mixed-signal integration is achieved within the flow. Tool installation and environment setup of OpenLane, OpenROAD, and Magic VLSI also required significant effort. Interpreting the role of abstraction files in integrating the analog macro was another difficulty. In future work, we aim to independently run the complete flow and analyze timing and area reports. Further study will focus on evaluating the impact of design parameter changes on overall implementation results.

REFERENCES

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